

Laboration 2

Förnamn Efternamn

2026-10-06

Sammanfattning

I detta inledande stycke ska ni förklara vad laborationen går ut på och vad ni kom fram till, utan att gå in på detaljer.

Uppgift 1

Teoretiska uppgifter

Uppgift 1.1

$$[a, a + (b-a)/k], [b - (b-a)/k, b]$$

Uppgift 1.2

$$(b-a)/k$$

Uppgift 1.3

$1/k$ $1 - 1/k$ Bernoulli fördelning med parametern $1/k$.

Uppgift 1.4

$$P(X_1 = j) = \binom{n}{j} \left(\frac{1}{k}\right)^j \left(1 - \frac{1}{k}\right)^{n-j} \quad X_1 \sim B(n, 1/k)$$

Uppgift 1.5

Ja, X_2, X_3 , osv. har samma fördelningen som X_1 . De är ej obereonde av varandra. Om till exempel $X_1 = n$, då medför det att $X_j = 0$ för alla andra $j \neq 1$.

Uppgift 1.6

$$E(X_1) = n/k \quad \text{Var}(X_1) = (n/k)(1 - 1/k)$$

$$E(Y_1) = E(X_1/n) = (1/n)E(X_1) = (1/n)(n/k) = 1/k. \quad \text{Var}(Y_1) = \text{Var}(X_1/n) = (1/n^2)\text{Var}(X_1) = (1/n^2)(n/k)(1 - 1/k) = (1/nk)(1 - 1/k). \quad D(Y_1) = \sqrt{\text{Var}(Y_1)} = \sqrt{(1/nk)(1 - 1/k)}.$$

Kodrelaterade uppgifter

Ännu mer text.

```
set.seed(19981207)

n <- 100
k <- 10

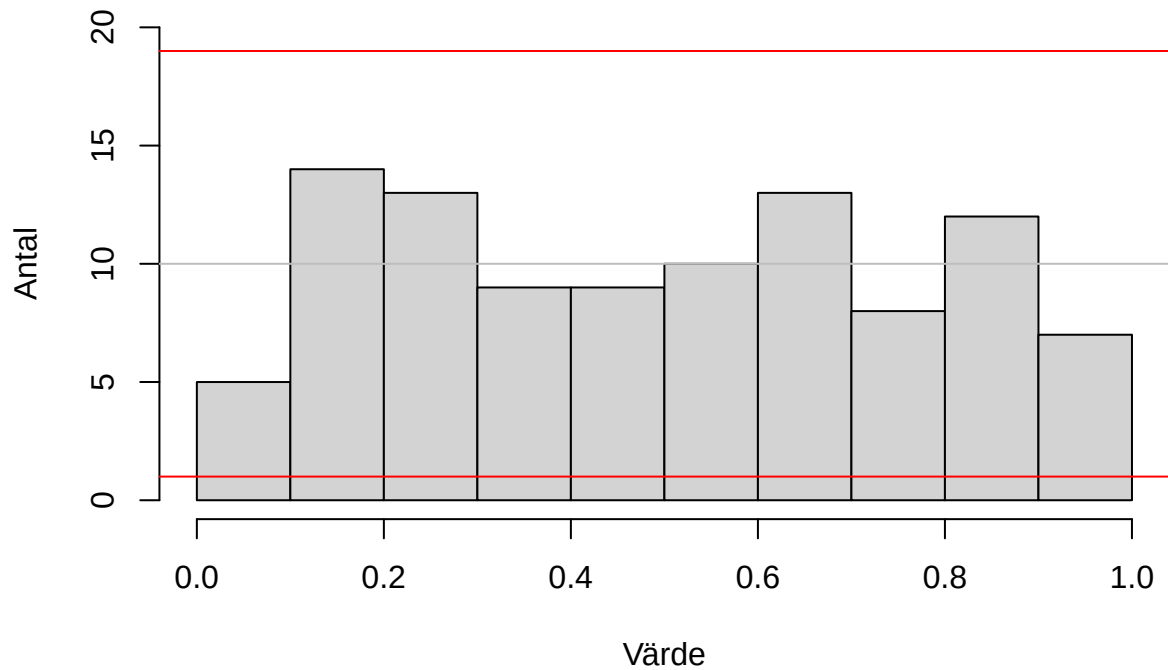
slumptal <- runif(n)

E_X = n/k
Var_X = (n/k)*(1 - 1/k)
D_X = sqrt(Var_X)

E_Y = E_X/n
Var_Y = Var_X/(n^2)
D_Y = sqrt(Var_Y)

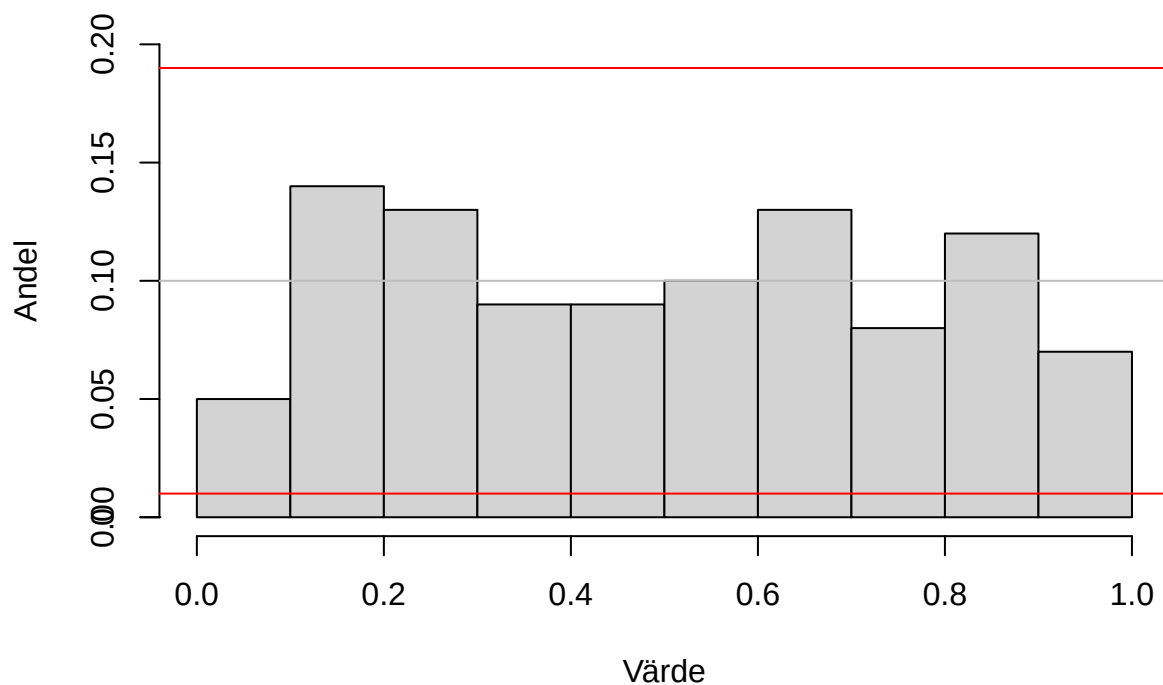
hist(slumptal,
      breaks = seq(from = 0, to = 1, length.out = k + 1),    # Skapar 10 delintervall
      main = "Histogram",
      ylab = "Antal",
      xlab = "Värde",
      ylim = c(0, 2*n/k))
abline(a = E_X, b = 0, col = "grey")
abline(a = E_X + 3 * D_X, b = 0, col = "red")
abline(a = E_X - 3 * D_X, b = 0, col = "red")
```

Histogram



```
prop_hist <- function(x, xlab = "Värde") {  
  p <- hist(x, breaks = seq(from = 0, to = 1, length.out = k + 1), plot = FALSE)  
  p$counts <- p$counts / sum(p$counts) # Konvertera antal till andelar  
  plot(p,  
    main = "Andelshistogram",  
    ylab = "Andel",  
    xlab = xlab,  
    ylim = c(0, 0.2))  
  axis(2)  
}  
  
prop_hist(slumptal)  
abline(a = E_Y, b = 0, col = "grey")  
abline(a = E_Y + 3 * D_Y, b = 0, col = "red")  
abline(a = E_Y - 3 * D_Y, b = 0, col = "red")
```

Andelshistogram



```
set.seed(19981207)

n <- 1000000
k <- 10

slumptal <- runif(n)

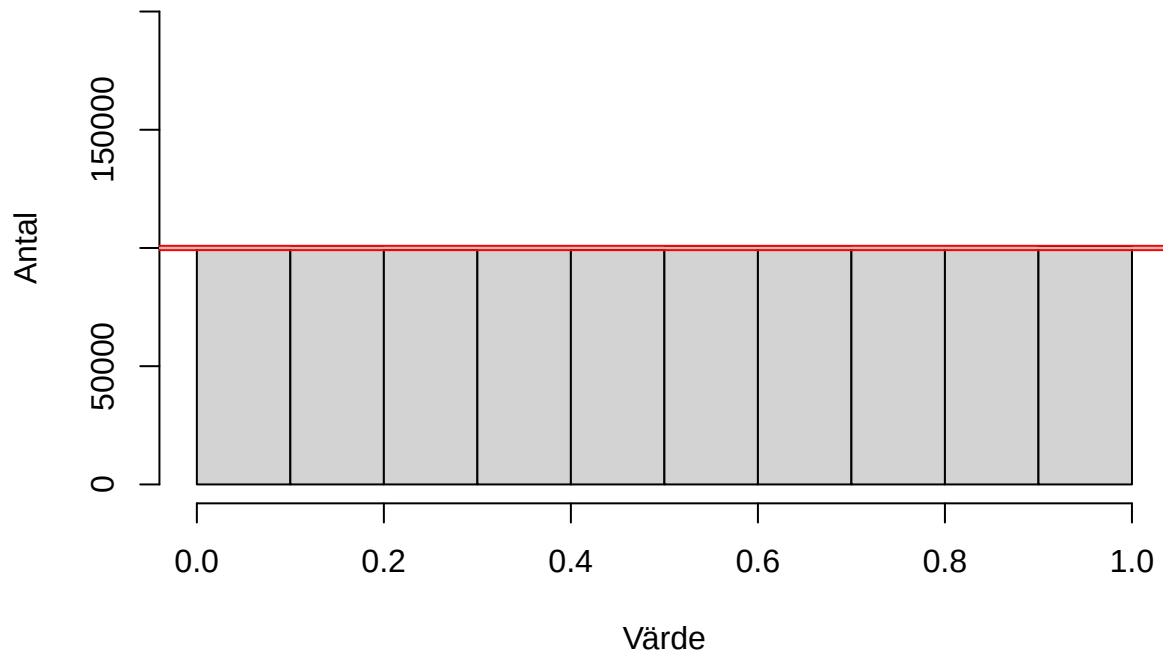
E_X = n/k
Var_X = (n/k)*(1 - 1/k)
D_X = sqrt(Var_X)

E_Y = E_X/n
Var_Y = Var_X/(n^2)
D_Y = sqrt(Var_Y)

hist(slumptal,
     breaks = seq(from = 0, to = 1, length.out = k + 1), # Skapar 10 delintervall
     main = "Histogram",
     ylab = "Antal",
     xlab = "Värde",
     ylim = c(0, 2*n/k))
abline(a = E_X, b = 0, col = "grey")
```

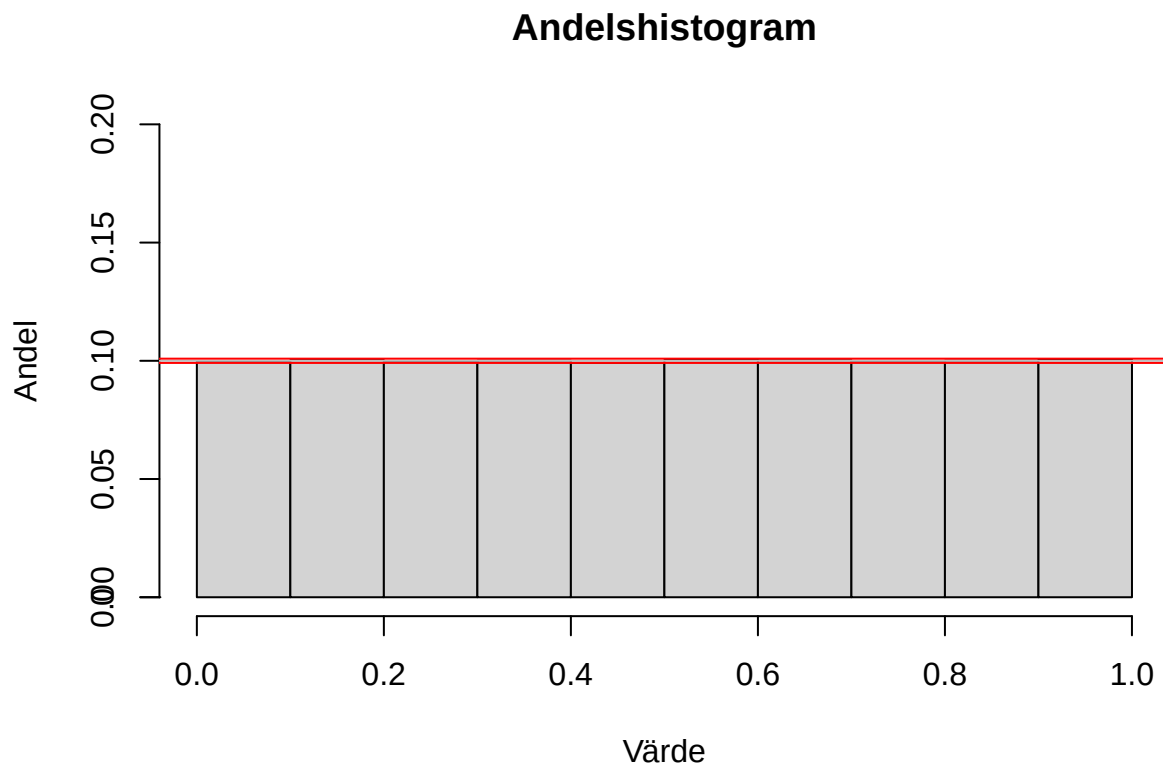
```
abline(a = E_X + 3 * D_X, b = 0, col = "red")
abline(a = E_X - 3 * D_X, b = 0, col = "red")
```

Histogram



```
prop_hist <- function(x, xlab = "Värde") {
  p <- hist(x, breaks = seq(from = 0, to = 1, length.out = k + 1), plot = FALSE)
  p$counts <- p$counts / sum(p$counts) # Konvertera antal till andelar
  plot(p,
        main = "Andelshistogram",
        ylab = "Andel",
        xlab = xlab,
        ylim = c(0, 0.2))
  axis(2)
}

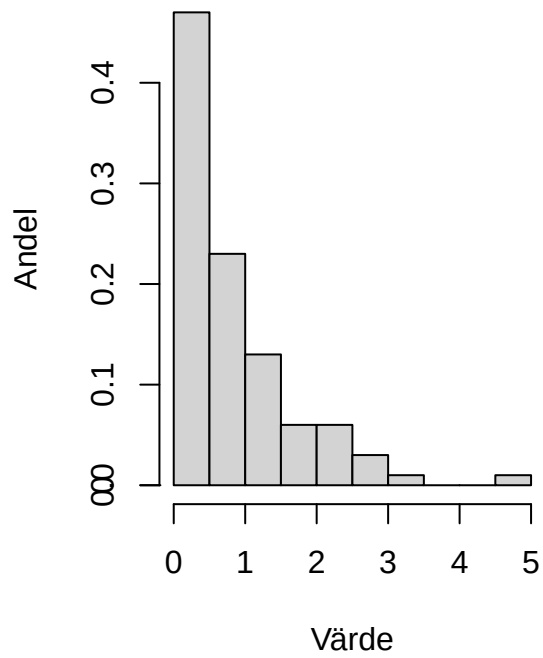
prop_hist(slumptal)
abline(a = E_Y, b = 0, col = "grey")
abline(a = E_Y + 3 * D_Y, b = 0, col = "red")
abline(a = E_Y - 3 * D_Y, b = 0, col = "red")
```



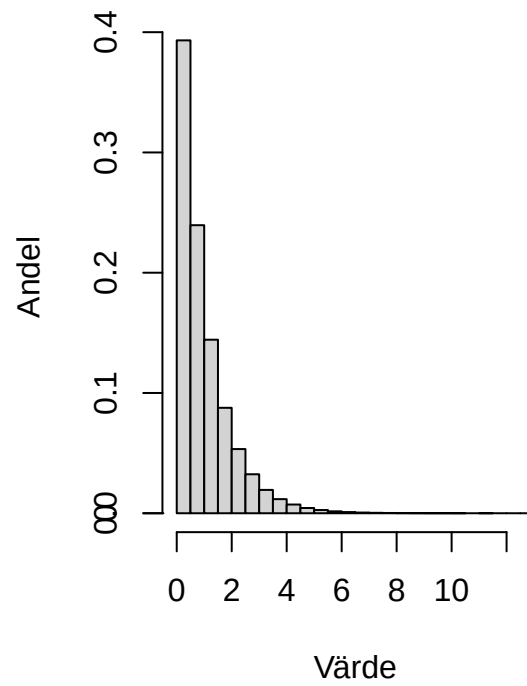
Uppgift 2

```
prop_hist <- function(x, xlab = "Värde") {  
  p <- hist(x, plot = FALSE)  
  p$counts <- p$counts / sum(p$counts)    # Konvertera antal till andelar  
  plot(p,  
    main = "Andelshistogram",  
    ylab = "Andel",  
    xlab = xlab)  
  axis(2)  
}  
n1 <- 100  
n2 <- 1000000  
a <- 1  
  
par(mfrow = c(1,2))  
prop_hist(rexp(n1, rate = a))  
prop_hist(rexp(n2, rate = a))
```

Andelshistogram

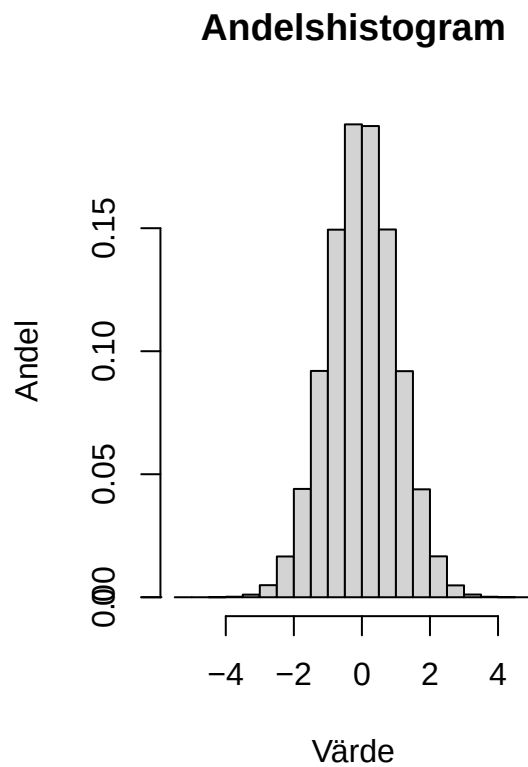
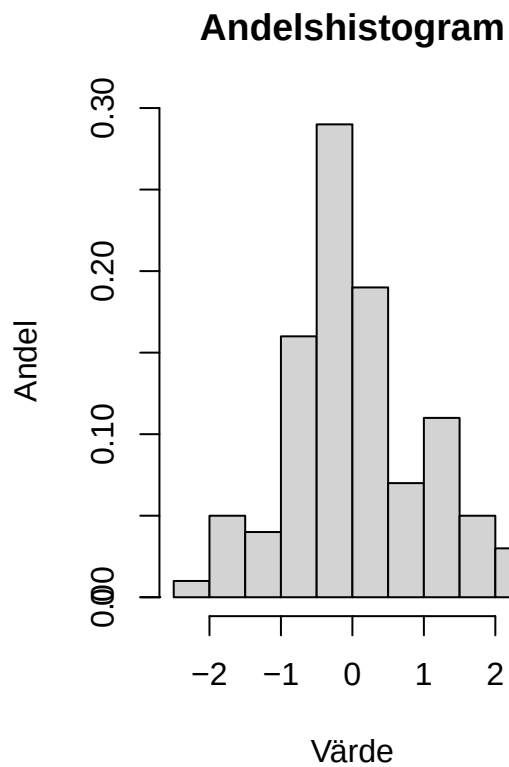


Andelshistogram



```
m <- 0
s <- 1

par(mfrow = c(1,2))
prop_hist(rnorm(n1, mean = m, sd = s))
prop_hist(rnorm(n2, mean = m, sd = s))
```



Uppgift 3

```
n <- 100
k <- 10

E_X = n/k
Var_X = (n/k)*(1 - 1/k)
D_X = sqrt(Var_X)

E_Y = E_X/n
Var_Y = Var_X/(n^2)
D_Y = sqrt(Var_Y)

x <- rexp(n, rate = 2)
v <- 1 - exp(-2 * x)

prop_hist <- function(x, xlab = "Värde") {
  p <- hist(x, plot = FALSE)
```

```

p$counts <- p$counts / sum(p$counts)    # Konvertera antal till andelar
plot(p,
      main = "Andelshistogram",
      ylab = "Andel",
      xlab = xlab,
      ylim = c(0, 0.2))
axis(2)
}

print(E_Y)

```

```
## [1] 0.1
```

```
print(D_Y)
```

```
## [1] 0.03
```

```

# Plotta x och v bredvid varandra
par(mfrow = c(1, 2))
prop_hist(x, xlab = "X")
prop_hist(v, xlab = "V")

abline(a = E_Y, b = 0, col = "grey")
abline(a = E_Y + 3 * D_Y, b = 0, col = "red")
abline(a = E_Y - 3 * D_Y, b = 0, col = "red")

```

